MATH 110

Dani Nguyen

Spring 2026

Contents

1

1.1 Lecture 1 - Introduction to PDEs

What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t),\ t \in \mathbb{R}$

$$\frac{du}{dt} = u(1-u) \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{du}{dt} = \mathbf{u} \quad \text{is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\begin{aligned} \text{Notation: } u_x &= \frac{\partial u}{\partial x} = \partial_x u \\ u_y &= \frac{\partial u}{\partial y} = \partial_y u \\ &\text{etc.} \end{aligned}$$

<u>Examples of PDEs</u>

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x, y)$

$$u_{xx} + u_{yy} = 0$$

$\boxed{\text{ex}}$ Consider the heat flow in a bar of length l

- $u(x, t)$ is the temperature of the bar at position x, time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = k u_{xx}$ where $t \geq 0, 0 \leq x \leq l$, k constant.

We also have:

- Boundary conditions (BCs):

$$u(0, t) = u(l, t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x,0) = \phi(x)$$

Remarks

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2^{nd} order
- A general 2^{nd} order PDE for $u(x,t)$ is:

$$G(x,t,u,u_x,u_t,u_{xx},u_{tt},u_{xt}) = 0$$

- A PDE in the above form is linear if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is homogeneous if every term contains u or some derivative of u. It is nonhomogeneous/inhomogeneous otherwise.

$\boxed{\text{ex}}$ Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2 t} u_{xt} = 0$
   is a linear and homogeneous PDE.
2. $u_t + 3xu_{xx} = \frac{t}{x}$
   is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.
3. $u_t + uu_x = 0$
   is a nonlinear PDE because of the uu_x term.
4. $u_{tt} + u_x + sin(u) = 0$
   is nonlinear because of the $sin(u)$ term.

Operating notation & linearity
We can write the heat equation $u_t - ku_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x,t)$ to produce a new function.

$$\begin{aligned} L(u) = Lu &= (\delta_t - k\partial_x^2)u \\ &= u_t - ku_{xx} \end{aligned}$$

L is a <u>linear</u> operator iff

$$\begin{aligned} L(u+v) &= Lu + Lv \\ &\qquad \forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ &\quad \forall \text{ constants } c \in \mathbb{R} \\ &\quad \forall \text{ functions } u \end{aligned}$$

$\boxed{\text{ex}}$ Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u+v) &= (\partial_t + k\partial_x^2)(u+v) \\ &= u_t + ku_{xx} + v_t + kv_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + cku_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

<u>Remarks</u>

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2 t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is
  - homogeneous if $f = 0$
  - inhomogeneous if $f \neq 0$
- A linear, homogeneous PDE $Lu = 0$ has the superposition principle

- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b.

$$\begin{aligned} Lu_1 &= Lu_2 = 0 \\ \Rightarrow L(au_1 + bu_2) &= 0 \end{aligned}$$

- The distinction between linear & nonlinear PDEs is very important.

<u>Solutions to PDEs</u>

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

$\boxed{\text{ex}}$ Consider the heat equation $u_t - u_{xx} = 0$.

$$\begin{aligned} u_1(x,t) &= x^2 + 2t \text{ is a soln} \\ \to \partial_t u_1 - \partial_x^2 u_1 &= 2 - 2 = 0 \\ u_2(x,t) &= e^{-t}\sin x \text{ is a soln} \\ \to \partial_t u_2 - \partial_x^2 u_2 &= -e^{-t}\sin x + e^{-t}\sin x = 0 \end{aligned}$$

$\boxed{\text{ex}}$ Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$\begin{aligned} Lu = f \text{ with } L &= \partial_x \\ f(x,t) &= tx \end{aligned}$$

We can solve this by direct integration on both sides wrt x.

$$u(x,t) = \frac{1}{2}tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

$\boxed{\text{ex}}$ Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$\begin{aligned} Lu &= f \\ \to L &= (\partial_t^2 - 4) \\ f(x,t) &= 0 \end{aligned}$$

There is no explicit x-dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x,t) = A(x)e^{-2t} + B(x)e^{2t}$$

1.2 Lecture 2 - Characteristics

Recap:

- Heat equation for $u(x,t)$

$$u_t - ku_{xx} = 0, \quad k > 0, \text{constant}$$

- We can write the heat equation as $Lu = 0$, where $L = (\partial_t - k\partial_x^2)$ is a differential operator.
- L is linear $\leftrightarrow$ $L(u+v) = Lu + Lv$ and $L(cu) = cLu$ $\forall$ functions u, v, constant c
- PDE: $Lu = f$ is linear if L is linear
- Linear PDE: $Lu = f$ is homogeneous if $f = 0$, inhomogeneous otherwise.

First-order linear PDE
Consider functions $u(x,y)$

$$u : \mathbb{R}^2 \to \mathbb{R}$$

Often we think of x, y as spatial variable and t as time variables. The simplest PDE we can have following this is:

$$u_x = 0$$

Which is 1st-order, linear, homogeneous. Also note that there is no explicit y-dependence. Therefore we can treat this like an ODE, and solve with direct integration of both sides.

$$u(x,y) = f(y)$$

where $f(y)$ is an arbitrary function of y.

Now consider $au_x + bu_y = 0$, which is a 1st-order, linear, homogeneous PDE with constants $a, b \in \mathbb{R}$.

$$\begin{aligned} \text{Let } V &= (a,b) \in \mathbb{R}^2 \\ \text{Recall } \nabla u &= (u_x, u_y) \\ \text{Then } V \cdot \nabla u &= (a,b) \cdot (u_x, u_y) \\ &= au_x + bu_y = 0 \end{aligned}$$

$\Rightarrow$ Directional derivative in direction V is 0.
$\Rightarrow$ u is constant on lines parallel to V.

Let $V^{\perp} = (b, -a)$, so that $V \cdot V^{\perp} = 0$.
Lines parallel to V satisfy

$$\begin{aligned} V^{\perp} \cdot (x,y) &= \text{constant} \\ \Rightarrow bx - ay &= \text{constant} \end{aligned}$$

Then u is constant on lines parallel to V.

$$\begin{aligned} \Rightarrow u(x,y) &= f(bx - ay) \\ \text{for an arbitrary function of } f \end{aligned}$$

<u>Check</u>:

$$\begin{aligned} &(a\partial_x + b\partial_y)\ f(bx - ay) \\ &= abf' - baf' = 0 \\ &\Rightarrow f' \text{ is a derivative of } f : \mathbb{R} \to \mathbb{R} \end{aligned}$$

Characteristic Coordinates. In this example, the lines parallel to V are called characteristics. Here u is constant along the characteristics.

We can write points (x, y) in characteristic coordinates (ξ, η). Where ξ is the x-intercept of the characteristic line, and η is the distance along that line to the point (x, y).

$$
\begin{aligned}
\text{So } x &= \xi + a\eta = x(\xi,\eta) \\
y &= b\eta = y(\xi,\eta) \\
\xi &= x - a\eta \\
\eta &= \frac{y}{b} \\
\xi &= x - \frac{a}{b}y \\
b\xi &= bx - ay
\end{aligned}
$$

Transform PDE to (ξ, η) coordinates.

Chain rule:

$$
\begin{aligned}
\frac{\partial u}{\partial \eta} &= \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \eta} \\
\Rightarrow u_\eta &= au_x + bu_y = 0
\end{aligned}
$$

Using characteristic coordinates (ξ, η), we've transformed to the PDE $u_\eta = 0$ for $u(\xi, \eta)$.

<u>Solution</u>:

$$
\begin{aligned}
u &= f(\xi) \\
\text{where } \xi &= \frac{1}{b}(bx - ay) \\
\Rightarrow u &= g(bx - ay)
\end{aligned}
$$

Where g is an arbitrary function.

<u>Remarks</u>:

- η is coordinates along characteristics
- u is constant on characteristics $\Rightarrow u_\eta = 0$
- ξ labels different characteristics. There are other choices for ξ but any choice $\tilde{\xi}$ would have $\tilde{\xi} = h(\xi)$

Method of Characteristics

Method for solving PDEs of form

$$a(x,y)u_x + b(x,y)u_y = 0 \quad \text{for } u(x,y)$$

Idea: Let $A(x,y) = (a(x,y), b(x,y))$

The directional derivative $A \cdot \nabla u = 0$

$\Rightarrow u$ is constant along curves w/ slope A.

- These are characteristic curves.
- Find coord η along chars
- Find coord ξ which labels chars
- PDE transforms to $u_\eta = 0$, general soln: $u = f(\xi)$

$\boxed{\text{ex}}$ $u_x + yu_y = 0$

The directional derivative along $(1, y)$ are 0.

The chars $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = 1,\ \frac{dy}{d\eta} = y$$

At $\eta = 0$, take $(x(0), y(0)) = (0, \xi)$

Solve:

$$\begin{aligned} \frac{dx}{d\eta} &= 1,\ x(0) = 0 \\ \Rightarrow x &= \eta \\ \frac{dy}{d\eta} &= y,\ y(0) = \xi \\ \Rightarrow y &= \xi e^{\eta} \end{aligned}$$

This follows that $\xi = ye^{-x}$.

$$\begin{aligned} \Rightarrow u_\eta &= u_x \frac{\partial x}{\partial \eta} + u_y \frac{\partial y}{\partial \eta} \\ &= u_x + \xi e^{\eta} u_y \\ &= u_x + yu_y = 0 \\ \Rightarrow u_\eta &= 0 \\ \Rightarrow u &= f(\xi) = f(ye^{-x}) \text{ with arbitrary function } f \end{aligned}$$

General solution:
Suppose we are given $u(0, y) = y^2$

$$\Rightarrow \text{Specific solution}$$
$$u(0, y) = f(y) = y^2$$
$$\Rightarrow u(x, y) = (ye^{-x})^2$$

Alternatively, from parametric equations,

$$\frac{dx}{d\eta} = 1 \qquad \frac{dy}{d\eta} = y$$
$$\Rightarrow \frac{dy}{dx} = y$$
$$\Rightarrow y = Ce^x$$

$\Rightarrow u$ is constant along chars.
$\Rightarrow u(x, Ce^x) = u(0, C)$
$= u(0, ye^{-x})$
$= f(e^{-x}y)$ for arbitrary f

$\boxed{\text{ex}}$ $u_x + 2xy^2u_y = 0$
Characteristic curves: $(x(\eta), y(\eta))$

$$\frac{dx}{d\eta} = 1 \qquad \frac{dy}{d\eta} = 2xy^2$$
$$\frac{dy}{dx} = 2xy^2$$

$$\int \frac{1}{y^2}dy = \int 2xdx$$
$$-\frac{1}{y} = x^2 - c$$
$$y = \frac{1}{c - x^2}$$

u is constant along chars.
$\Rightarrow u(x, y) = f(c) = f(x^2 + \frac{1}{y})$ is the general soln.

Advection, transport

Consider $u(x,t)$ describing quantity u at position x, time t.

$$u_t + cu_x = 0 \quad (c > 0)$$

Chars: $(t(\eta), x(\eta))$ satisfy

$$\begin{aligned} &\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = c \\ &\Rightarrow \frac{dx}{dt} = c \\ &\Rightarrow x = ct + \xi \ \text{ via integration} \end{aligned}$$

General solution:

$$u = f(\xi) = f(x - ct)$$

1.3 Lecture 3 - Conservation Laws

Recap:

- Solve PDEs of the form $a(x,y)u_x + b(x,y)u_y = 0$
- The char curves $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = a(x,y) \qquad \frac{dy}{d\eta} = b(x,y)$$

- u is constant along chars.

$\boxed{\text{ex}}$ Given $\frac{1}{x^2}u_x + u_y = 0$

$$\frac{dx}{d\eta} = \frac{1}{x^2} \qquad \frac{dy}{d\eta} = 1$$

Method 1: divide to get parametric eq.

$$\begin{aligned} \frac{dy}{dx} &= x^2 \\ y &= \frac{1}{3}x^3 + \xi \end{aligned}$$

where constant ξ labels the char.

The general solution is $u = f(\xi) = f(y - \frac{1}{3}x^3)$ for an arbitrary function f. This can be confirmed by plugging u back into the PDE.

Method 2: solve parametric equations.

$$\frac{dx}{d\eta} = \frac{1}{x^2} \qquad \frac{dy}{d\eta} = 1$$
$$\Rightarrow \int x^2 dx = d\eta \qquad y = \eta + B$$
$$\Rightarrow \frac{1}{3}x^3 = \eta + A \qquad y = \eta + B$$

We can absorb the two constants into one ξ

$$\Rightarrow \frac{1}{3}x^3 = \eta + \xi \qquad y = \eta$$
$$\Rightarrow y = \frac{1}{3}x^3 - \xi$$

We should check that the chars cover the whole plane, i.e. no chars have been 'missed', via sketching or finding

$$\eta = \eta(x, y)$$
$$\xi = \xi(x, y)$$

Coord transformation is also useful

$$\begin{matrix} \frac{1}{3}x^3 = \eta + \xi \\ y = \eta \end{matrix} \Rightarrow \begin{matrix} x = (3\eta + 3\xi)^{\frac{1}{3}} \\ y = \eta \end{matrix}$$

This transform our PDE

$$\frac{1}{x^2}u_x + u_y = 0 \iff u_\eta = 0$$
$$\text{Gen soln: } u = f(\xi)$$
$$= f(\frac{1}{3}x^3 - y)$$

with arbitrary function f

$\boxed{\text{ex}}$ Advection
Consider function $u(x,t)$ describing a quantity u at position x and time t

$$u_t + cu_x = 0 \tag{Advection}$$

with constant $c > 0$
Chars $(t(\eta), x(\eta))$ satisfy

$$\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = c$$

Solve

$$\begin{aligned} t &= \eta + A \\ x &= c\eta + B \end{aligned}$$

Absorb into one constant using $A = 0, B = \xi$

$$\begin{aligned} \Rightarrow t &= \eta \\ x &= c\eta + \xi \\ \Rightarrow x &= ct + \xi \end{aligned}$$

By inverting, $\eta = t,\ \xi = x - ct$.
Now we have an transformed PDE

$$u_\eta = u_t + cu_x = 0$$

General solution is

$$\begin{aligned} u(x,t) &= f(\xi) \\ &= f(x - ct) \end{aligned}$$

for an arbitrary f.

Suppose we had initial condition $u(x,o) = \phi(x)$, then $u(x,0) = f(x) = \phi(x) \Rightarrow$ solution is $u(x,t) = \phi(a - ct)$.

<u>Remarks</u>:
We can solve more complicated PDEs using chars, for example,

$$u_t + cu_x = f(u)$$

Transform to coords (ξ, η) with $t = \eta$, $x = c\eta + \xi$

$$u\eta = f(u)$$

which can be solved as an ODE

Conservation Laws. (Logan 1.2)

- Where do PDEs come from?
- For a conserved quantity:

$$\begin{matrix}\text{rate of change} \\ \text{in domain}\end{matrix} \implies \begin{matrix}\text{net flow into domain} \\ + \text{ creation in domain}\end{matrix}$$

e.g. Animal population in a fixed region

Quantitatively

- Let $u(x,t)$ be density of some quantity in 1d domain.
- The amount of quantity in section dx is $u(x,t)Adx$.
- Let $\phi(x,t)$ be the flux, which is the amount of quantity crossing section x at time t.
- $A\phi(x,t)$ is amount of quantity passing x at time t
- Let $f(x,t)$ be the rate at which quantity is created.
- $f(x,t)Adx$ is amount of quantity created in width dx per unit time.

Remarks:

- $\phi(x,t) > 0 \implies$ flux is L to R
- $\phi(x,t) < 0 \implies$ flux is R to L
- $f(x,t) > 0$ is source at x (creation)
- $f(x,t) < 0$ is sink at x (destroyed)

Balance:

$$\begin{matrix}\text{change in}\\ \text{quantity in}\\ \text{section}[a,b]\end{matrix} = \text{flux} + \text{creation}$$

$$\Rightarrow \frac{d}{dt}\int_a^b u(x,t)A dx = A\phi(a,t) - A\phi(b,t) + \int_a^b f(x,t)A dx$$

$$\Rightarrow \int_a^b u_t A dx = -\int_a^b \phi_x A dx + \int_a^b f A dx$$

$$\Rightarrow \int_a^b (u_t + \phi_x - f)dx = 0$$

This is true for any domain $[a,b]$

$$u_t + \phi = f \text{ where}$$

$$\begin{matrix}\text{change in}\\ \text{quantity}\end{matrix} = \text{net flux} + \text{net creation}$$

Remarks:
Can get a PDE for u by

- Specifying $f(x,t)$
- Choosing a relationship between ϕ and u, u_x, etc. This is called constitutive law.
- E.g. $\phi = cu \rightarrow$ advection.

Diffusion:

- Conservation law with no sources

$$u_t + \phi_x = 0$$

- Suppose $u(x,t)$ is density of a gas of particles. Then
  - Particles move from high u to low u.

- Flux increase with density gradient.
- Fick's law:

$$\phi(x,t) = -Du_x$$

where $D > 0$ is diffusion constant.
- This gives the diffusion equation

$$u_t = Du_{xx}$$

which is the same as the heat equation.

Remark:

- Can combine diffusion with other effects.
e.g. Advection-diffusion eq.

$$\text{flux: } \phi = cu - Du_x$$

Balance:

$$\begin{aligned} u_t + \phi_x &= 0 \\ u_t + cu_x &= Du_x x \end{aligned}$$

1.4 Lecture 4 - Diffusion Equation

Recap:

- Conservation Laws (Logan 1.2)
- Density: $u(x,t)$
- Flux: $\phi(x,t)$ (movement of u)
- Sources: $f(x,t)$ (creation of u)
- Balance in domain $[a,b]$

$$\frac{d}{dt}\int_a^b Au(x,t)dx = A\phi(a,t) - A\phi(b,t) + \int_a^b Af(x,t)dx$$

$$\int_a^b u_t dx = -\int_a^b \phi_x dx + \int_a^b f dx$$

$$\int_a^b (u_t + \phi_x - f)dx = 0$$

This holds for any $[a, b]$

$$u_t + \phi_x = f$$
$$\begin{matrix}\text{change in}\\ \text{quantity}\end{matrix} + \text{flux} = \text{source}$$

Diffusion equation.
Suppose $u(x, t)$ is density/concentration of substance. We have Fick's Law for flux:

$$\phi(x, t) = -Du_x,\ D > 0$$

- Movement from high u to low u
- Steeper gradient gives higher flux

Conservation/balance Law:

$$\begin{aligned} u_t + \phi_x &= 0 \\ \Rightarrow u_t &= Du_{xx} \end{aligned} \qquad \text{(Diffusion/heat eq)}$$

<u>Remarks</u>:

- D is called diffusivity or thermal conductivity.
- Some physical settings have a non-constant diffusivity $D(x)$

  $$\phi = -D(x)u_x$$

  This yields the PDE

  $$\begin{aligned} u_t &= (Du_x)_x \\ &= D'u_x + Du_{xx} \end{aligned}$$
  $$\text{where } D' = \frac{dD}{dx}$$

- Can also find nonlinear diffusion equations where $D = D(u)$

  $$u_t = (D(u)u_x)_x$$

- We can combine diffusion with other terms, e.g. diffusion-advection

$$u_t + cu_x = Du_{xx}$$
$$\phi(x,t) = -Du_x + cu$$

<u>Dimensions/units:</u>
Suppose $u(x,t)$ is temperature, then

$$[u] = \text{temp}$$
$$[u_t] = \frac{\text{temp}}{\text{time}}$$
$$[u_{xx}] = \frac{\text{temp}}{\text{length}^2}$$

So $u_t = Du_{xx}$

$$\Rightarrow [D] = \frac{\text{length}^2}{\text{time}}$$

Consider heat flow in a region of length L. The time-scale T for temperature change is

$$T = \frac{L^2}{D} \text{ so } D = \frac{L^2}{T}$$

Boundary & Initial Conditions. Diffusion eqs usually have BCs and ICs. Consider heat flow in 1d with domain $0 < x < l$.

IC has form

$$u(x,0) = u_0(x)$$

where $u_0(x)$ is initial temperature profile.

There are 3 types of common BCs.

1. Dirichlet BCs
   Specify u at the boundary.

$$u(0,t) = g_1(t)$$
$$u(l,t) = g_2(t)$$

   where g_1, g_2 are given i.e. u is known at boundary.

2. Neumann BCs
   Specify u_x at the boundary.

$$\begin{aligned} u_x(0,t) &= h_1(t) \\ u_x(l,t) &= h_2(t) \end{aligned}$$

   where h_1, h_2 are given, i.e. flux is known at boundary.

3. Robin BCs
   Specify u, u_x given at boundary.

$$\begin{aligned} au(0,t) + bu_x(0,t) &= \psi_1(t) \\ au(l,t) + bu_x(l,t) &= \psi_2(t) \end{aligned}$$

   where ψ_1, ψ_2 are given. E.g. Newton's Law of Cooling states that heat flow $\propto$ temp difference between object and environment.

$$\begin{aligned} bu_x(0,t) &= -au(0,t) + \psi_1(t) \\ \text{flux} &= \text{object} + \text{environment} \end{aligned}$$

<u>Remarks</u>:

- A well-posed problem is a PDE with ICs and BCs s.t. it has a unique solution with a correct physical meaning (stability).
- The above terminology for BCs applies to general PDEs.

Steady-state solutions. Many time-dependent PDEs have solutions which approach steady states $u = u(x)$, which are time-independent. E.g. diffusion/heat equation:

$$u_t = Du_{xx}$$

Steady state solution $u(x)$ has $u_t = 0$, so

$$Du'' = 0 \quad (u' = u_x)$$

This gives a 2nd-order ODE, where

$$u(x) = Ax + B$$

for constants A, B set by BCs.

$\boxed{\text{ex}}$ Consider the diffusion-decay equation

$$u_t = Du_{xx} - ru$$
$$0 < x < L,\ t, D, r > 0$$

u has the flux $\phi = -Du_x$ and is destroyed at rate ru. Take BCs:

$$u(0,t) = 0$$
$$-Du_x(L,t) = -1$$
$$t > 0$$

and IC

$$u(x,0) = g(x)$$
$$0 < x < L$$

<u>Remarks</u>:
This model could describe a diffusing pollutant (Du_{xx}) which decays at rate ru.

$$\begin{aligned} \text{At } 0: & \ u(0,t) = 0 \\ & \Rightarrow u \text{ is held at } 0 \\ \text{At } L: & -Du_x(L,t) = -1 \\ & \Rightarrow \text{constant flux of } u \text{ into domain} \end{aligned}$$

Steady state problem:

$$u = u(x),\ \text{so } u_t = 0$$

BVP:

$$Du'' - ru = 0$$
$$0 < x < L$$

BCs: $u(0) = 0,\ u'(L) = \frac{1}{D}$
Here we ignore IC and assume transients have decayed.
Solution of BVP:

$$u'' = \frac{r}{D}u$$
$$u(0) = 0,\ u'(L) = \frac{1}{D}$$

General solution:

$$u = A\sinh\left(x\sqrt{\frac{r}{D}}\right) + B\cosh\left(x\sqrt{\frac{r}{D}}\right)$$
$$\text{consts } A, B$$

BC1:

$$\begin{aligned} u(0) &= 0 \\ \Rightarrow B &= 0 \\ \Rightarrow u(x) &= A\sinh\left(x\sqrt{\frac{r}{D}}\right) \end{aligned}$$

BC2:

$$\begin{aligned} u'(L) &= \frac{1}{D} \\ \Rightarrow A\sqrt{\frac{r}{D}}\cosh\left(L\sqrt{\frac{r}{D}}\right) &= \frac{1}{D} \\ \Rightarrow u(x) &= \frac{\sinh\left(x\sqrt{\frac{r}{D}}\right)}{\sqrt{rD}\cosh\left(L\sqrt{\frac{r}{D}}\right)} \end{aligned}$$

<u>Remarks</u>:

- An ODE BVP may not have a solution. Physically the PDE may not have a steady state solution.
- The steady state solution could be unstable in general. (Here it turns out that $\lim_{t\to\infty} u(x,t) \to$ steady state.)

Summary. Diffusion in 1d, for $u(x,t)$

$$\begin{aligned} u_t &= Du_{xx} \\ D &> 0 \end{aligned}$$

Can consider higher dimensions for $u(x,y,t)$

<u>Preview</u>: Vibrations and waves
Consider the 1d wave equation for $u(x,t)$:

$$u_{tt} = C^2 u_{xx}$$

This describes, for example, waves on a string. C has dimensions length/time and is the speed of waves. 2nd-order in time $\rightarrow$ waves (contrasts with 1st-order in time $\rightarrow$ diffusion).

We have the string fixed at length $x = 0, L$ and $u(x,t)$ is the height of the string at time t. And we have the following equation of motion for the string

$$u_{tt} = \sqrt{\frac{T}{\rho}} u_{xx}$$

where T is string tension, and ρ is the density of string.

1.5 Lecture 5 - Wave Equation